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Per- and polyfluoroalkyl substances (PFAS) inhibit cytochrome P450 CYP3A7 through direct coordination to the heme iron and water displacement

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Abstract

Per- and polyfluoroalkyl substances (PFAS) are a chemical class of highly stable, fluorinated compounds popular for use in a variety of consumer products. PFAS environmental persistence in drinking water contributes to acute exposure in humans and subsequent bioaccumulation of the compounds in the liver and lung tissue. Prenatal PFAS exposure has been associated with lowered birth weight, premature birth, and developmental defects including cranio-facial abnormalities. The cytochrome P450 enzyme CYP3A7 is responsible for facilitating a variety of reactions essential for proper fetal development in humans. In addition to drug metabolism, CYP3A7 is responsible for metabolizing endogenous ligands in the developing human liver, including the steroid precursor dehydroepiandrosterone 3-sulfate (DHEA-S), essential for estriol synthesis during pregnancy, along with the morphogen all-*trans*-retinoic acid (*at*RA). Interference with estriol synthesis during pregnancy, as well as *at*RA clearance, is known to result in similar effects associated with prenatal PFAS exposure including lowered birth weight, premature birth, and developmental defects. We hypothesized that PFAS compounds bind to the CYP3A7 enzyme resulting in its inhibition. We implemented a series of binding studies using spectral characterization of six PFAS compounds (PFOA, PFOS, GenX, PFNA, PFNS, and PFHxS), and evaluated their interactions with recombinant CYP3A7. In addition, we screened PFAS for their

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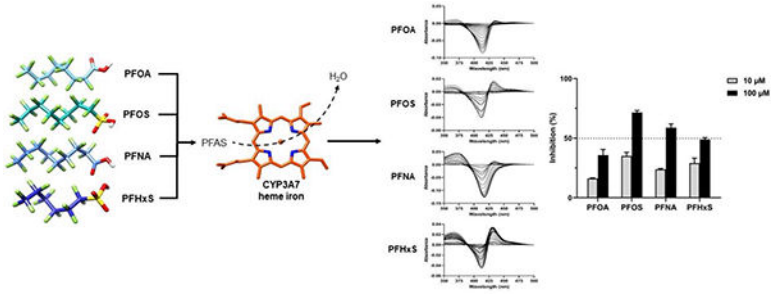
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ability to inhibit CYP3A7 oxidative activity using dibenzylfluorescein, a fluorescent probe, and DHEA-S, an endogenous substrate of CYP3A7. Our data demonstrate that of the six PFAS tested, PFOA, PFOS, PFNA, and PFHxS bind to and inhibit CYP3A7.

Graphical Abstract:



Per- and polyfluoroalkyl substances (PFAS) coordinate with the heme iron in the CYP3A7 active site and displace water resulting in both reverse Type I and Type I binding. These unique interactions feature negative cooperativity related to PFAS concentration and propensity for electrostatic repulsion, resulting in inhibition of CYP3A7 oxidative activity.

Keywords

Per- and polyfluoroalkyl substances (PFAS); Cytochrome P450; CYP3A7; Developmental toxicity; Type I and reverse Type I ligand binding; Enzyme kinetics

1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are a chemical class of over 4,700 fluorinated aliphatic compounds known for their oleophobic and hydrophobic qualities [1,2]. Their physicochemical stability makes them popular for use in a vast multitude of consumer products including disposable food and toy packaging, clothing, carpeting, and cookware [3,4]. PFAS are also major components of aqueous film-forming foams (AFFF), which belong to a category of flame retardants often used at airports and military bases for training purposes [4,5]. These compounds can leach or shed from consumer products into food, air, and soil leading to background human exposure [4,6]. In addition, runoff from manufacturing plants, airports and military bases into local waterways serves as the main source for chronic PFAS exposure in humans, as we currently lack the proper infrastructure to filter these environmentally persistent compounds from consumer water [5,7].

Their remarkable resilience to degradation pathways has been attributed to the strong carbon-fluorine (C-F) bonds in PFAS alkyl chains, which explains their widespread prevalence in human tissues with half-lives reaching 5 years [8,9]. Legacy compounds with longer alkyl chains such as perfluorooctanoic acid (PFOA) and perfluorosulfonic acid (PFOS) have been manufactured since the 1940s and have been thoroughly studied in the literature [10]. Findings of associations between these compounds and adverse health effects has led to their gradual ban in signatory countries by the Stockholm Convention

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on Persistent Organic Pollutants (POPs) [11]. These legacy compounds have since been replaced by novel shorter-chain compounds such as hexafluoropropylene oxide-dimer acid (HFPO-DA, or GenX) with six carbons in its alkyl chain (Figure 1). Although it has been argued that substitutes for legacy compounds are more biologically inert, it has been observed in recent animal models that these novel compounds possess similar or higher toxic potency than their predecessors [12]. Legacy compounds remain ubiquitous in nearly 98% of human serum in the United States and maintain their relevancy in the scientific literature despite their phase-out [13,14]. Further research is necessary to compare the biochemical properties and subsequent health effects of the two classes, to either confirm or deny these claims.

PFAS exposure in humans is associated with a variety of health effects including immunosuppression, as well as endocrine and metabolic disruption [15,16]. First exposure to PFAS can start as early as the *in-utero* period via maternal umbilical cord blood. In the human embryo, these compounds have been found to bioaccumulate in the lungs during the first trimester of development and in the liver during the second and third trimesters of development [17]. PFAS can additionally be transferred from mother to infant through breast milk after birth, and formula-fed infants remain exposed through their intake of public drinking water [18,19]. Prenatal PFAS exposure is associated with preterm birth, lowered birth weight, and developmental defects such as craniofacial abnormalities [20,21,22].

The human hepatic cytochrome P450 (CYP) enzyme CYP3A7 comprises nearly half the CYP content in the neonatal and fetal liver [23,24]. CYP3A7 is expressed primarily during fetal development and up to 12 months after birth performing many of the same functions as its adult isoform CYP3A4, with which it shares 87% of its genetic sequence [25,26]. In addition to drug metabolism, CYP3A7 is responsible for metabolism of endogenous compounds such as dehydroepiandrostenedione 3-sulfate (DHEA-S) which it hydroxylates into 16α-hydroxy-DHEA-S [27]. This is a key step in the estriol biosynthetic pathway during pregnancy. Estriol synthesis up-regulates uterine vasodilation and is necessary to bring a pregnancy to full-term [28,29]. In addition to metabolism of DHEA-S, CYP3A7 is one of the CYP enzymes responsible for clearance of all-*trans*-retinoic acid (*at*RA), a metabolite of vitamin A1, from the developing liver [30]. Improper clearance of *at*RA results in accumulation and subsequent hepatotoxicity. High levels of *at*RA in the neonatal and fetal liver have been associated with teratogenicity resulting in developmental defects such as cranio-facial abnormalities, as well as spontaneous abortion [31,32].

The promiscuous substrate specificity of the CYP3A isoforms predispose them to interactions with ligand mimickers that share structural resemblance with endogenous ligands [33]. PFAS compounds not only possess long alkyl chains similar to those of fatty acids, but certain PFAS have identical functional groups to those of known CYP3A7 ligands (Figure 1). PFOS, perfluorononanesulfonic acid (PFNS), and perfluorohexane sulfonic acid (PFHxS), all prominent PFAS in the literature, possess sulfonic acid head groups such as that of DHEA-S. Additionally, PFOA, GenX, and perfluorononanoic acid (PFNA) each contain carboxylic acid head groups comparable to that of *at*RA (Figure 1). Given these chemical resemblances, there is a propensity in the CYP3A7 binding site for low-affinity interactions with these familiar functional groups found on PFAS.

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These potential interactions may lead to subsequent inhibition of CYP3A7 activity by PFAS compounds. Previous studies have shown PFAS inhibition of other CYP enzymes involved in steroid and lipid synthesis including CYP2C19, CYP2E1, and CYP2D6, as well as CYP3A4 [34]. However, interactions between PFAS and CYP3A7 are not well characterized. Given the similarities in the literature between the effects of prenatal PFAS exposure and those related to inhibition of estriol synthesis and *at*RA clearance, we hypothesized that PFAS compounds can bind to and inhibit CYP3A7.

We performed binding studies to determine the specific interactions between CYP3A7 and six prominent PFAS in the literature (PFOA, PFOS, GenX, PFNA, PFNS, and PFHxS). CYP enzymes largely display characteristic absorbance spectra which can be used to monitor ligand binding. The type of spectra represents the specific interactions that take place between a ligand and the heme iron in the binding pocket of the CYP enzyme. Typical CYP absorbance spectra are grouped into three different categories: Type I, reverse Type I, or Type II. Spectra are classified based on unique properties related to absorbance values at different wavelengths, and the overall shape of the curve. Type I and reverse Type I spectra are both indicative of a displacement of water by the ligand in the CYP's binding pocket. However, during reverse Type I binding, there is an additional interaction between a heteroatom from the ligand (e.g., oxygen or sulfur), with the heme iron in the binding site of the CYP. This results in spectra which mirror that of Type I spectra, reflected across the x-axis. Type II absorbance spectra occur as a result of a nitrogen atom on a ligand coordinating with the CYP's heme iron and forming a covalent bond. Given that none of the PFAS tested possess a nitrogen atom, we expected to observe either Type I or reverse Type I spectra when characterizing these interactions. Our observations were verified through molecular docking studies.

Additionally, the PFAS that were shown to interact with CYP3A7 were screened for their ability to inhibit the oxidative activity of the enzyme. The data obtained from these experiments provide insight into the biochemical mechanisms leading to developmental toxicity resulting from prenatal PFAS exposure.

2. MATERIALS AND METHODS

2.1 Chemicals.

Dimethyl sulfoxide (DMSO) was purchased from Alfa Aesar (Haverhill, MA). PFOA (CAS: 335–95-5) was purchased from Thermo Fisher Scientific Inc. (Waltham, MA). PFOS (CAS: 2795–39-3) was supplied by Sigma-Aldrich (St Louis, MO). GenX (CAS: 62037–80-3) was purchased from Manchester Organics (Runcorn, UK). PFNA (CAS: 375–95-1) and PFHxS (CAS: 3871–99-6) were both acquired through Frontier Scientific Inc. (Logan, UT). PFNS (CAS: 98789–57-2) was purchased from Toronto Research Chemicals (Toronto, ON). All PFAS stocks were prepared fresh in DMSO prior to each experiment (5–100 mM). Glucose-6-phosphate, glucose-6-phosphate dehydrogenase, β-nicotinamide adenine dinucleotide phosphate ($NADP^+$), and dehydroepiandrosterone-d_5-3-sulfate (DHEA-S-d5) were all purchased from Sigma Aldrich (St. Louis, MO). The fluorescent probe substrate, dibenzylfluorescein (DBF, CAS: 97744–44-0) and DHEA-S (CAS: 1099–87-2) were supplied by Cayman Chemical Company (Ann Arbor, MI). The metabolite standard, 16α-

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hydroxy DHEA-S, was purchased from Steraloids (Newport, RI). The CYP3A7 Bactosomes (catalog number: CYP/EZ060) were purchased from Cypex Ltd. (Scotland, UK) where the recombinant CYP3A7 enzyme was co-expressed with the human cytochrome P450 reductase in *E. coli* and then supplemented with the human cytochrome b_5.

2.2. Recombinant CYP3A7 Protein Expression.

Recombinant human CYP3A7 (rCYP3A7) was expressed and purified based on the protocol outlined by Godamudunage *et al.* [26]. An N-terminally truncated, codon-optimized, and His-tagged CYP3A7 cDNA cloned into the pCWori+ vector (pCWori-CYP3A7d4–24H4) was a generous gift from the Dr. Emily Scott Lab (Department of Medicinal Chemistry, University of Michigan, Ann Arbor, Michigan, USA). The construct was designed for expression in chemically competent *Escherichia coli* DH5α F'IQ cells (Thermo-Fisher Scientific, Waltham, MA) that have been previously transformed with the pGro7 chaperone plasmid (Emily Scott Lab). Transformation of *E. coli* DH5α/pGro7 cells with pCWori-CYP3A7d4–24H4 was achieved via heat shock and grown overnight at 37°C on Luria Broth (LB) plates containing chloramphenicol (20 μg/mL) and ampicillin (100 μg/mL). CYP3A7 was expressed under the control of the *tac* promoter of the pCWOri$^+$ plasmid using *E. coli*-DH5α F'IQ cells (Thermo-Fisher Scientific, Waltham, MA) in Terrific Broth (TB) medium. Briefly, 20 mL of an overnight pCWori-CYP3A7 *E. coli*-DH5α starter culture consisting of LB medium supplemented with 20 μg/mL chloramphenicol and 200 μg/mL ampicillin was used to inoculate each liter of TB (also containing 20 μg/mL chloramphenicol and 200 μg/mL ampicillin). The bacterial cultures were incubated in 2.8L PYREX Erlenmeyer Flasks (Charleroi, PA; No. 4420) at 37°C under agitation until the optical density reached an absorbance at 600 nm of 0.5. Upon reaching 0.5 OD_{600}, the GroES/EL chaperone expression was induced by adding arabinose (2 mg/mL) and 1 mM δ-aminolaevulinic acid was also added at that time. The incubation temperature was reduced to 25°C under agitation until reaching an OD_{600} of 0.6–0.8. At that optical density, CYP3A7 expression was induced by the addition of 1 mM isopropyl ß-D-1-thiogalactopyranoside (IPTG). Following induction, cells were grown for 48 h at 25°C under agitation, then were harvested via centrifugation at 2,400 × g at 4°C. Cells were further processed to spheroplasts by resuspending them in lysis buffer [20 mM TrisHCl (pH 8.0), 20% glycerol, 1 mM PMSF, 10 mM β -mercaptoethanol] while adding chicken egg white lysozyme (Millipore-Sigma, St. Louis, MO) at 0.3 mg/mL and stirring at 4°C for 30 min. Lysed cells were centrifuged for 15 min at 8,300 × g at 4°C, and the spheroplast pellet was collected and stored at −80°C until ready for purification of the expressed CYP3A7 enzyme.

2.3 Recombinant (His)-Tagged CYP3A7 Protein Purification.

Spheroplast pellets were thawed at room temperature and resuspended in cold buffer consisting of 20 mM TrisHCl (pH 8.0), 20% glycerol, 1 mM PMSF, 10 mM β-mercaptoethanol. The spheroplast suspension was then lysed on ice with a Branson sonicator set at 100% power and five 30-sec bursts with 30-sec resting time between each burst. The lysate was then centrifugated for 15 min at 8,300 × g and 4°C to remove spheroplast debris. The resulting supernatant underwent ultra-centrifugation for 45 min at 100,000 × *g* and 4°C. The pellet obtained was then homogenized in buffer consisting of 50 mM Tris-HCl pH 8, 100 mM NaCl, 1 mM PMSF, 5 mM β-mercaptoethanol, and 20% glycerol.

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To help the solubilization of CYP3A7, 1% 3-((3-cholamidopropyl) dimethylammonio)-1-propanesulfonate (CHAPS) was gradually added and stirred up to 2 h at 4°C. The mixture was finally ultra-centrifugated for 50 min at 100,000 × *g* and 4°C. The CYP3A7-containing supernatant was loaded onto a Ni-NTA HisTrap-HP column (Cytiva, Marlborough, MA) pre-equilibrated with the loading buffer consisting of 50 mM Tris-HCl pH 8, 500 mM NaCl, 10 mM imidazole, 10 mM CHAPS, and 20% glycerol. The column was subsequently washed with five column volumes (CV) of loading buffer, followed by 5 CV of wash buffer (100 mM potassium phosphate pH 7.4, 500 mM NaCl, 10 mM imidazole, 10 mM CHAPS, and 20% glycerol). rCYP3A7 was then eluted with a gradient of imidazole using the elution buffer of 10 mM potassium phosphate pH 7.4, 100 mM NaCl, 250 mM imidazole, 10 mM CHAPS. Red colored fractions were pooled and diluted 5-fold with the carboxymethyl (CM) wash buffer consisting of 5 mM potassium phosphate pH 7.4, 1 mM ethylenediaminetetraacetic acid (EDTA), 6 mM CHAPS, and 20% glycerol. The diluted pooled sample was loaded on a HiTrap CM sepharose FF column (Cytiva, Marlborough, MA) and then washed with ten column volumes of CM wash buffer. The rCYP3A7 protein was eluted based with a 10 CV salt-gradient using the elution buffer of 50 mM potassium phosphate, pH 7.4, 500 mM NaCl, 1 mM EDTA, and 20% glycerol. The CM sepharose fractions containing the bulk of the rCYP3A7 protein were pooled, concentrated, and dialyzed at 4°C in 100 mM potassium phosphate pH 7.4, 1 mM EDTA, 0.1 mM dithiothreitol (DTT), and 20% glycerol. The final protein concentration of the purified rCYP3A7 was determined by bicinchoninic acid assay (Pierce, Thermo Fisher Scientific). The final concentration of the ferrous-CO protein was determined using UV–visible spectroscopy on an Agilent Cary 60 spectrophotometer (Agilent Technologies; Santa Clara, CA), with an extinction coefficient of $\varepsilon = 91\ mM^{-1}\ cm^{-1}$ at the wavelength of 450 nm [35].

2.4 Optical Difference Spectroscopy of PFAS Binding to rCYP3A7.

UV-visible difference spectroscopy studies were performed on an Agilent Cary 60 spectrophotometer (Agilent Technologies; Santa Clara, CA). Binding of PFOA [10 to 600 μM], PFOS [2.5 to 200 μM], GenX [2.5 to 600 μM], PFNA [10 to 700 μM], PFNS [2.5 to 300 μM], and PFHxS [50 to 1500 μM] to CYP3A7 were tested (n=3). Positive controls for Type I (Lopinavir [0.25 to 20 μM]) and Type II (Ketoconazole [0.25 to 150 μM]) binding were acquired for comparison (n=1). Concentration ranges used were optimized based on data from preliminary screening. All stock solutions were prepared and diluted by serial dilution in DMSO. Both sample and reference chambers contained 1 mL of rCYP3A7 (2 μM for PFAS; 1 μM for controls) in 100 mM potassium phosphate (pH 7.4). Prior to initiating the titration, a baseline was recorded between 350 and 500 nm. Aliquots of ligand solutions were added to the sample cuvette, whereas equal volume of DMSO was added to the reference cuvette to determine the difference spectrum at varying PFAS concentrations. The absolute changes in absorbance deriving from triplicate titrations were plotted as a function of ligand concentration. The dissociation constant (K_d), R^2, and B_{max} for each ligand was calculated with the nonlinear regression Hill (*h*) equation (equation 1) using GraphPad Prism 9 for MacOS (version 9.4.1) (San Diego, CA).

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$$Y = B\max * X^h/(Kd^h + X^h) \quad (1)$$

Where Y is the fraction of the receptor protein (CYP) concentration that is bound by the ligand, X is the total ligand concentration, B_{max} is the absolute spectral response, and h is the Hill coefficient.

2.5 In-Silico Docking Studies of PFOA, PFOS, PFNA, and PFHxS to CYP3A7.

Molecular docking studies were performed for the four PFAS compounds binding to CYP3A7: PFOA, PFOS, PFNA, and PFHxS using AutoDock Vina v. 1.1.2 [36]. Studies were performed using the experimental crystal dimer structure of fetal-specific CYP3A7 (PDB: 7MK8) from Sevrioukova *et al.* [37] downloaded from the Protein Data Bank (PDB: https://www.rcsb.org/). The receptor was prepared for docking by removing ions and water molecules and adding polar hydrogens using MGL AutoDock Tools v 1.5.7 (UCSD Molecular Graphics Lab and The Scripps Research Institute). PFAS 3D structures were downloaded from PubChem (https://pubchem.ncbi.nlm.nih.gov/), converted to PDB, and parameterized by adding polar hydrogens and identifying rotatable bonds. Coordinates for the receptor docking grid search space were established by defining the enzyme active site, with the following parameters: grid box center: x-center = 93.937, y-center = 284.197, z-center = 174.386, and total number of grid points in each direction: x-dimension = y-dimension = z-dimension = 30. A configuration docking script was prepared using MS Notepad in text format with the exhaustiveness search parameter set to 24. AutoDock Vina was executed using the configuration file with PDBQT.out and log.out file options included. The PDBQT.out file contains all binding poses for any particular docking simulation. Output files were analyzed using the ViewDock feature in UCSF Chimera v. 1.15 [38]. Docking energies and associated distances (Å) between each heteroatom (reverse Type I) or ω−1 carbon (Type I) interaction for each PFAS were recorded (Table 2). Relevant docking poses were selected based on energetic favorability and proximity to the heme iron.

2.6 PFAS in vitro Inhibition Screening Assay for Recombinant CYP3A7 using the DBF Probe Substrate.

PFAS compounds which exhibited successful binding to rCYP3A7 were evaluated for their capacity to inhibit CYP3A7 bactosome activity using DBF as a fluorescent probe substrate. CYP3A7 bactosomes were implemented instead of purified rCYP3A7 to more faithfully recapitulate kinetics *in vivo*. Experimental setup was based on the DBF high-throughput assay previously described in Work *et al.* [39]. In brief, PFOA, PFOS, PFNA, and PFHxS stocks (5 and 50 mM) were prepared fresh daily in DMSO and combined with the DBF-recombinant CYP3A7 (0.15 μM DBF; 10 pmol/mL CYP3A7 bactosomes) incubations done in 0.1 M potassium phosphate buffer (pH 7.4) and 3.3 mM magnesium chloride to yield a 10 and 100 μM PFAS final concentrations, respectively. DMSO was used for solvent control and did not exceed 0.2% final concentration across reactions to minimize non-specific CYP interactions. Reactions (final volume of 100 μL), in triplicate, were initiated at 37°C by the addition of the NADPH-regenerating system mix consisting of $NADP^+$ (1 mM), glucose-6-phosphate (10 mM), and glucose-6-phosphate dehydrogenase (2 IU/mL) and were stopped by the addition of 75 μL 2 M NaOH. After 1 h incubation of the stopped samples

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at 37°C, formation of the fluorescent metabolite of DBF, fluorescein, was measured using a Tecan Infinite M Plex microplate reader (Männedorf, Switzerland) to determine CYP3A7 activity. Additional incubations without the NADPH-regenerating system mix were done in parallel as negative controls. The solvent control samples were used as reference for 100% activity to calculate the percent inhibition in samples containing the PFAS compounds. The GraphPad Prism 9 software for MacOS (version 9.4.1) was used for data processing, graph generation and descriptive statistics (San Diego, CA).

2.7 PFAS IC_{50} Inhibition Assays for Recombinant CYP3A7 and DBF Oxidation.

PFAS compounds that achieved over 50% inhibition in the CYP3A7 inhibition screening assay (PFOS and PFNA) were further characterized for their inhibition properties by carrying out IC_{50} assays. Similar experimental procedure as the screening assay was followed. PFOS (80 mM) and PFNA (150 mM) stock solutions were prepared in DMSO and working solutions were made by serial dilution in DMSO to yield final PFOS and PFNA concentrations in incubations ranging from 1.25 to 160 μM and 2.5 to 300 μM, respectively. DMSO was used as the solvent control and did not exceed 0.2% final concentration across reactions to prevent non-specific CYP interactions. Percent remaining activities with increasing PFAS concentrations were calculated based on the initial velocity of the formation of the DBF metabolite, fluorescein, against the solvent control. The half maximal inhibitory concentrations (IC_{50}) for each PFAS were determined based on the dose-response curve fitting using the log(inhibitor) vs. response – variable slope (equation 2) from GraphPad Prism 9 software for MacOS (version 9.4.1) (San Diego, CA).

$$Y = Bottom + (Top - Bottom)/(1 + 10\hat{}((LogIC50 - X) * HillSlope)) \quad (2)$$

2.8 PFOS IC_{50} Inhibition Assay for Recombinant CYP3A7 and DHEA-S Hydroxylation.

Since PFOS had the lowest IC_{50} value for CYP3A7 DBF oxidation, it was further characterized for its *in vitro* inhibition potency for CYP3A7 DHEA-S hydroxylation. The IC_{50} assay was performed according to the procedure previously outlined in Kandel et al. [27] with the following modifications: PFOS stock and working solutions were prepared in DMSO at the concentrations previously described for the DBF oxidation assay and CYP3A7 Bactosome reactions (10 pmol/mL) were incubated up to 5 min at 37°C. After removal of precipitated proteins by centrifugation, supernatant samples were analyzed by LC-MS/MS for formation of the 16α-hydroxy DHEA-S metabolite according to the analytical method described in Kandel et al. [27] and using the Xevo TQ-S micro tandem quadrupole mass spectrometer (Waters Corp. Milford, MA) with multiple reaction monitoring (MRM) scan type. The MS peaks were integrated using QuanLynx software (version 4.1, Waters Corp., Milford, MA) and the analyte/internal standard peak area ratios were used for relative quantification. The mean analyte/internal standard peak area ratio for 16α-hydroxy DHEA-S was determined for the solvent control and was referred as the 100% control activity to calculate the percent remaining activity in samples containing increasing concentrations of PFOS. The GraphPad Prism software was used for dose-response curve fitting.

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3. RESULTS:

3.1 PFAS Unique Binding Interactions with CYP3A7.

Of the six PFAS tested, PFOA, PFOS, PFNA, and PFHxS triggered unique changes in the heme Soret spectrum upon binding to the recombinant CYP3A7. These interactions featured two distinct binding events at similar wavelengths consistent across all four PFAS: reverse Type I binding [40,41] (λ 430 – 436 nm), and Type I binding [40,41] (λ 365 – 371 nm); each representative peak in the spectra share a trough (λ 410 – 415 nm) as well (Figure 2) (Table 1). This spectral behavior demonstrated negative cooperativity [42] with the reverse Type I binding initially observed during the titration, but then a shift occurred wherein Type I binding occurred as the enzyme became more saturated with each PFAS compound. This inverse relationship favors the first reverse Type I binding event which is demonstrated by lower K_d values for the initial interaction compared to that of the second Type I binding event.

PFOA [2.5 – 600 μM] demonstrated its initial reverse Type I binding event at λ = 436 nm, K_d = 237.5 μM (R^2= 0.975), and its second Type I binding event at λ = 370 nm, K_d = 477.9 μM (R^2= 0.990). The first binding event for PFOS [2.5 – 200 μM] occurred at λ = 434 nm, K_d = 61.74 μM (R^2= 0.970), and its second binding event occurred at λ = 365 nm, K_d = 90.39 μM (R^2= 0.973). PFNA [30 – 700 μM] exhibited its reverse Type I behavior at λ = 435 nm, K_d = 76.87 μM (R^2= 0.894), and its Type I binding at λ = 371 nm, K_d = 181.2 μM (R^2= 0.923). PFHxS [50 – 1500 μM] demonstrated its initial binding event at λ = 430 nm, K_d = 296.8 μM (R^2= 0.977), and its Type I binding at λ = 365 nm, K_d = 731.3 μM (R^2= 0.962) (Figure 2) (Table 1).

The PFAS compounds with carboxylic acid head groups (PFOA and PFNA) demonstrated binding peaks and troughs at proportionately higher wavelengths for both binding events compared to PFAS compounds with sulfonic acid head groups (PFOS and PFHxS) (Figure 2). In addition, longer chain length, and possession of a sulfonic acid head group appeared to result in more effective binding, as evidenced by the lower K_d values observed (Table 1). GenX and PFNS did not demonstrate any spectral interactions with rCYP3A7 in these experiments.

Positive controls for Type I (Lopinavir) and Type II (Ketoconazole) binding demonstrated respective K_d values of 6.482 μM (R^2 = 0.992) and 23.24 μM (R^2 = 0.997) (Table 1, Figure S1).

3.2 In-silico Docking Studies Provide Structural Support for the Type I and Reverse Type I PFAS-CYP3A7 Interactions.

PFOA, PFOS, PFNA, and PFHxS all demonstrated energetically favorable docking poses consistent with reverse Type I and Type I interactions with the CYP3A7 heme iron observed in our previous binding experiments (Figure 3). Reverse Type I interactions featured the closest distances (Å), and lowest binding energies (kcal/mol) between the oxygen heteroatom of each PFAS compound and the heme iron: PFOA (Figure 3, panel A: 3.427 Å; −7.7 kcal/mol), PFOS (Figure 3, panel C: 3.149 Å; −7.9 kcal/mol), PFNA (Figure 3, panel E: 3.400 Å; −8.2 kcal/mol) and PFHxS (Figure 3, panel G: 3.078 Å; −7.2

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kcal/mol) (q.f. Table 2). These configurations involved the PFAS of interest docking in a perpendicular fashion to the heme iron with the hydrophilic head groups oriented towards the heme and the alkyl chain tail pointing away from the center of the active site. Docking poses representative of a Type I interaction between the ω−1 carbon of each PFAS and the heme iron consisted of slightly longer distances between the PFAS molecule ω−1 carbon and the heme iron when compared to the distances retrieved from the PFAS oxygen hetero atom. In addition, the heme iron in the reverse Type I pose indicated higher binding energies compared to those of the Type I poses observed for all the PFAS compounds examined: PFOA (Figure 3, panel A: 4.346 Å; −8.3 kcal/mol), PFOS (Figure 3, panel C: 5.818 Å; −8.0 kcal/mol), PFNA (Figure 3, panel E: 4.054 Å; −8.3 kcal/mol) and PFHxS (Figure 3, panel G: 4.279 Å; −7.5 kcal/mol) (q.v. Table 2). Type I binding configurations involved each PFAS docking parallel to the heme iron with the ω−1 carbon oriented above the heme, and the hydrophilic head groups pointed in the opposite direction away from the heme iron.

3.3 PFAS Inhibition of CYP3A7 Oxidative Activity.

All four PFAS compounds that bound to CYP3A7 (PFOA, PFOS, PFNA, and PFHxS) also demonstrated inhibition of the enzyme in the DBF oxidation assay [39] when tested at both 10 and 100 μM concentrations (Figure 4). PFOS was observed to be the most potent inhibitor at both concentrations tested, whereas PFOA was the least potent. Percent inhibition (± SD) of CYP3A7 DBF oxidation at 10 μM PFOS was 35.1% ± 2.99 and 71.5% ± 1.66 at 100 μM PFOS. Inhibitory potency of PFOS was followed by PFNA at 23.9% ± 0.593 (10 μM PFNA) and 58.8% ± 3.15 (100 μM PFNA) along with PFHxS at 29.1% ± 4.19 (10 μM PFHxS) and 48.8% ± 1.63 (100 μM PFHxS). Our least potent inhibitor, PFOA, inhibited the reaction at 16.3% ± 0.275 (10 μM PFOA) and 35.8% ± 4.76 (100 μM PFOA) percent inhibition of CYP3A7 activity (Figure 4). This data was consistent with our findings from the PFAS binding studies with recombinant CYP3A7.

3.4 Quantitative Determination of the Extent of Inhibition of PFAS and PFNA CYP3A7 DBF Oxidation.

IC_{50} values were determined for two of the most potent inhibitors of CYP3A7 DBF oxidation, PFOS and PFNA (Figure 5). As the IC_{50} data illustrates, PFOS had the lowest half maximal inhibitory concentration with an IC_{50} value of 20.39 μM (R^2 = 0.981) compared to that of PFNA, which had an IC_{50} of 137.6 μM (R^2 = 0.978) (Figure 5, A–B). This data is consistent with that of our PFAS inhibition screening and binding studies.

3.5 Quantitative Determination of the Inhibition Extent of PFOS for CYP3A7 DHEA-S Hydroxylation.

The *in vitro* inhibitory potency of PFOS, the most potent PFAS for the CYP3A7 DBF oxidation, was then assessed for the CYP3A7 16α-hydroxylation of the adrenal steroid hormone DHEA-S (Figure 6). DHEA-S, at the K_m concentration of 5 μM, was incubated with the CYP3A7 bactosomes and a concentration range of PFOS. An IC_{50} value of 16.75 μM (R^2 = 0.980) was estimated which was consistent with that of the DBF oxidation assay (Figure 5).

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4. DISCUSSION:

In recent years, the class of compounds collectively known as PFAS has garnered a significant amount of attention due to their widespread environmental contamination and the subsequent lingering health effects caused by human exposure to them. These include certain types of cancer, hepatotoxicity, and immune dysfunction, as well as birth and developmental defects. While clear associations between these compounds and their adverse effects on human health exist, the underlying molecular toxicological mechanisms have remained elusive. From a chemical perspective they bear a structural resemblance to fatty acids, although their preponderance of C-F bonds make them unlike any naturally occurring fatty acid-type molecules, while contributing to their unique physiochemical properties. Since the C-F bond does not occur in the biological context, it is reasonable to assume that it is unlikely that these compounds would undergo significant metabolism by fatty acid hydroxylases or CYP enzymes. However, their ultra-hydrophobic nature would suggest that they may be effective *competitive inhibitors* of these enzymes, given the hydrophobicity of CYP active sites. Since CYP enzymes are involved in a significant number of critical biological processes, including regulation of the immune system (through arachidonic acid oxidation), development of cancer, and fetal growth and development, this inhibition could have profound implications for the human pathologies observed with these compounds.

Based on our findings presented in this study, it is clear that PFAS compounds can bind to (Figures 2,3) and inhibit (Figures 4,5) CYP3A7 function *in vitro*. The unique nature of the absorption difference spectra demonstrates two distinct binding events between recombinant CYP3A7 and four of six PFAS tested (PFOA, PFOS, PFNA, and PFHxS) (Figure 2). This behavior is indicative of a negatively cooperative relationship which arises from the lower K_d values obtained for the second binding event across all four PFAS compounds when compared to those of the first binding event. Interestingly, negative cooperativity in ligand binding has been reported previously for CYP3A enzymes [42]. In Roberts *et al.*, it was observed that testosterone binds to CYP3A4 exhibiting negative cooperativity [42]. The authors concluded that an initial high-affinity testosterone binding event was followed by the binding of a second testosterone molecule with a lower-affinity, hence demonstrating negative cooperativity for this closely related enzyme.

Due to the spectral characteristics of the absorption difference spectra obtained from our ligand binding studies, we can deduce that the first binding event in each case is representative of reverse Type I binding, in which a negatively charged oxygen atom from the PFAS head group displaces the axial water ligand to form a coordinate covalent bond with the heme iron in the binding pocket of the CYP or, alternatively, the ligand forms a bond with the heme iron via a bridged water atom, as observed with certain substrates and inhibitors [43]. This interaction occurs at a lower K_d and PFAS concentration compared to the second binding event, indicating a higher affinity for this particular binding pose. As the PFAS concentration increases throughout the titration, the binding behavior switches to the more energetically favorable Type I binding mode, indicated by a separate peak arriving in our absorbance spectra at ~375 nm. As the initial reverse Type I complex becomes saturated, the PFAS binds in a parallel orientation to the heme, likely to avoid electrostatic repulsion

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between the negative heteroatoms in the ligand head groups. This, in turn, displaces the first molecule from coordination with the heme, resulting in a Type I spectrum.

Our docking simulations reinforce this interpretation of the data. As can be seen in Figure 3, panel A, one of the most energetically favorable poses for PFOA shows it in a perpendicular orientation in regard to the position of the carboxylic acid and the heme iron, with a distance of 3.247 Å between the alcohol oxygen and the heme iron, indicative of a reverse Type I interaction (distances of less than 5 Å are generally considered significant for an interaction). Another favorable binding pose indicated that PFOA could also bind in a horizontal orientation with respect to the heme (Figure 3, panel B), in this case with the hydrophobic end of the molecule within 4.346 Å of the heme iron. Similar docking poses were also obtained with PFOS, PFNA, and PFHxS (Figure 3, panels C-H, respectively), indicating a consistent mode of interaction between the different PFAS compounds and CYP3A7.

The significant differences in the B_{max} values point to the fact that the individual respective ligands have differential effects on the spin-state conversion of the heme iron (Figure 2). It has been noted previously that certain ligands may be more effective in promoting spin-state conversion of the heme iron to the high (or low) spin-state forms (Hlavica, P., 2007). In the case of the PFAS compounds, surprisingly the trend is that the weaker binding compounds (PFOA and PFHxS) produced the greatest B_{max}, indicating spin-state conversion for an increased portion of the enzyme (Figure 2). The only exception to this was the second binding event detected for PFNA, which produced the largest B_{max} (the B_{max} for the initial binding events for PFNA and PFHxS were similar). While PFOS has a lower half maximal inhibitory concentration than PFNA, the inhibition by PFOS is only decreased to ~30% activity of the control reaction, whereas PFNA reaches total inhibition (Figure 4). This could be related to observed difference in B_{max} values, as noted above (Figure 2).

The K_d values from our binding studies were consistent with our results from both the CYP3A7 inhibition and IC_{50} experiments. Of the PFAS tested, PFOS displayed the highest affinity for CYP3A7 through its interaction with the heme iron. In addition, PFOS was our most potent inhibitor of CYP3A7 oxidative activity, in which 10 μM [PFOS] resulted in 35.1% inhibition, and 100 μM [PFOS] achieved 71.5% inhibition. This potency is likely due to the presence of the sulfate group in the hydrophilic portion of the alkyl chain, which is an important binding moiety in the endogenous ligand of CYP3A7, DHEA-S. In contrast, the increased carbon chain length in PFNA resulted in a lower affinity than PFOA and similarly, a lower inhibition potency, allowing it to reach 23.9% inhibition at 10 μM [PFNA] and 58.8% inhibition at 100 μM [PFNA] (Figure 4). Given that PFNA has a lower affinity for rCYP3A7 compared to PFOS, and that PFNS did not bind to rCYP3A7, it is clear that alkyl chain length is not solely determinative of PFAS inhibitory potential of the enzyme. In terms of these interactions, it appears that PFAS compounds containing a sulfonic acid head group do not benefit from a chain length beyond 8-carbons, whereas compounds possessing a carboxylic head group can compensate for their weaker affinity for the heme iron through a longer alkyl chain that facilitates hydrophobic interactions. Further research is needed to understand how this unique paradox operates on a biochemical level.

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The initial observations we made of the K_d values in our spectral binding studies indicated low-affinity interactions between PFAS and the heme iron in the binding pocket of CYP3A7 compared to other ligands. For example, lopinavir and ketoconazole bound to CYP3A7 with K_d values of 6.482 μM (R^2 = 0.992) and 23.24 μM (R^2 = 0.997), respectively (Table 1, Figure S1). Lopinavir possesses a terminal elimination half-life of 2.33 to 2.44 hours depending on dosage frequency [Boffito et al., 2008, International Medical Press]. Ketoconazole has a biphasic half-life of about two hours for the first 10 hours following a dose, and then a terminal half-life of about 8 years [44]. Although serum PFAS concentrations present in the nanomolar range, PFAS half-lives can reach 5 years in human tissues [16,17,9,45]. In addition, both lopinavir and ketoconazole are active substrates of CYP enzymes, namely CYP3A4, and their mechanisms of clearance are well-established in the literature. It has not been established whether PFAS compounds can be metabolized in humans. The compounds are thought to be eliminated in the bile and urine, and are known to undergo enterohepatic recirculation, as well as become reabsorbed in renal tubules, allowing for increased biological permanence compared to other CYP3A7 inhibitors [46,47,48]. Thus, given their long half-lives and propensity to accumulate in the liver, the PFAS compounds examined here may be effective competitive inhibitors of endogenous CYP3A7 ligands, such as DHEA-S and *at*RA.

Indeed, PFOS showed significant inhibition potency for CYP3A7 *in vitro* 16α-hydroxylation of DHEA-S with an IC_{50} of 16.8 μM, in particular when considering that the K_m for DHEA-S is approximately 5 μM. Further research is needed to determine whether the PFAS compounds studied here have an impact on CYP3A7 oxidation of DHEA-S and *at*RA *in vivo*. Furthermore, testing PFAS interactions with other CYP enzymes involved in development would provide additional insight into the specific biochemical implications of exposure.

Overall, our data demonstrate unique inhibitory interactions between prominent PFAS, and the important fetal cytochrome P450 enzyme CYP3A7. Future efforts may shed light on specific *in vivo* biochemical mechanisms of toxicity of this important class of environmental pollutants.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Elsevier Highlights

- PFAS compounds are toxicants that can inhibit cytochrome P450 (CYP) enzymes
- Certain PFAS bind to CYP3A7 induce both Type I and reverse Type I spectra
- The binding is best fit to a model of negative cooperativity
- PFAS can inhibit CYP3A7 oxidation of DBF
- Inhibition of CYP3A7 by PFAS could help explain their developmental toxicity

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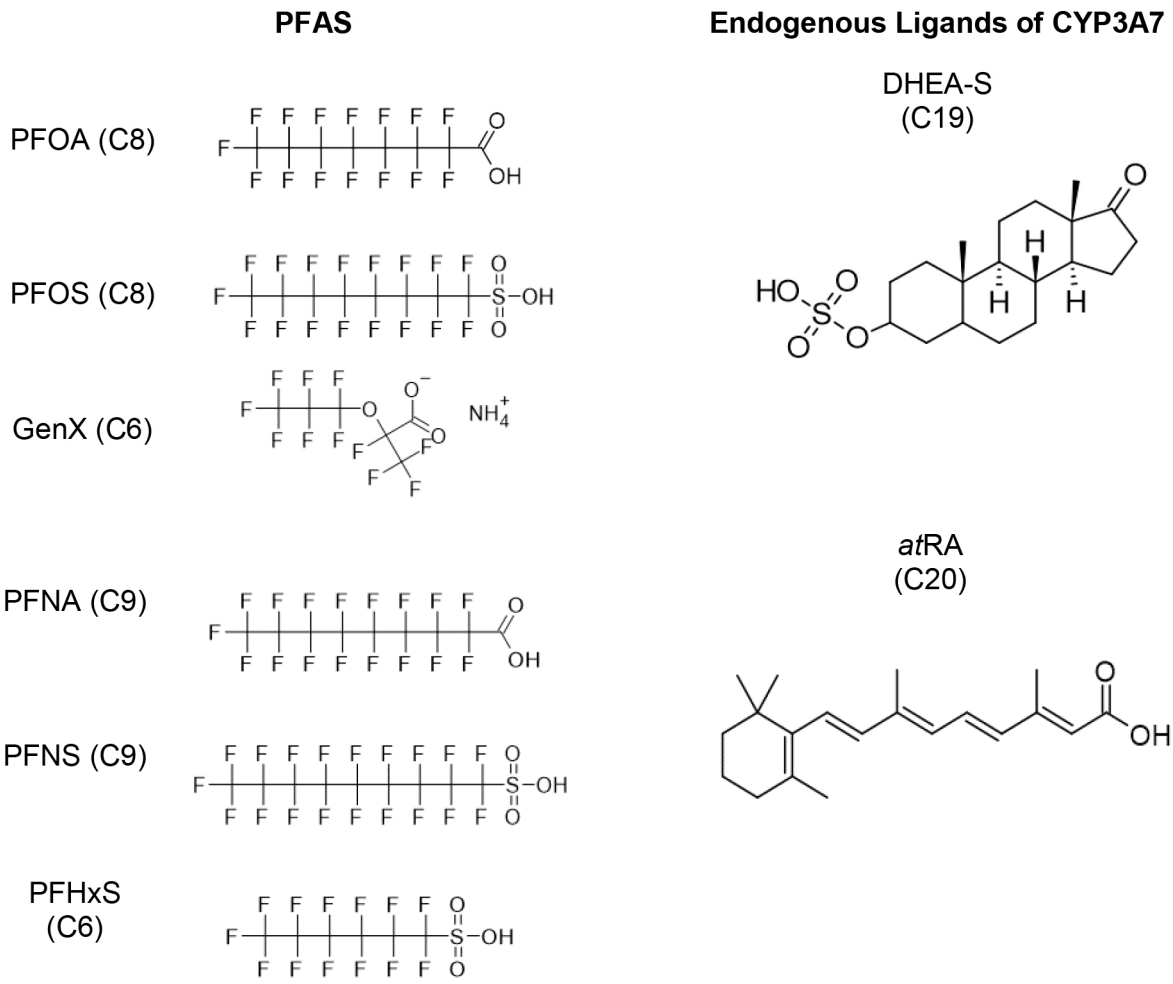


Figure 1. Chemical structures of the six prominent PFAS compounds tested and two endogenous ligands of CYP3A7.

Compound names with the number of carbons are provided in parentheses for each structure. PFOA, perfluorooctanoic acid; PFOS, perfluorooctanesulfonic acid; GenX (HFPO-DA), hexafluoropropylene oxide-dimer acid; PFNA, perfluorononanoic acid; PFNS, perfluorononanesulfonic acid; PFHxS, perfluorohexanesulfonic acid; DHEA-S, dehydroepiandrostenedione 3-sulfate; *at*RA, all-*trans*-retinoic acid. Structures were drawn using the ACD/ChemSketch software (Freeware) 2021.2.1.

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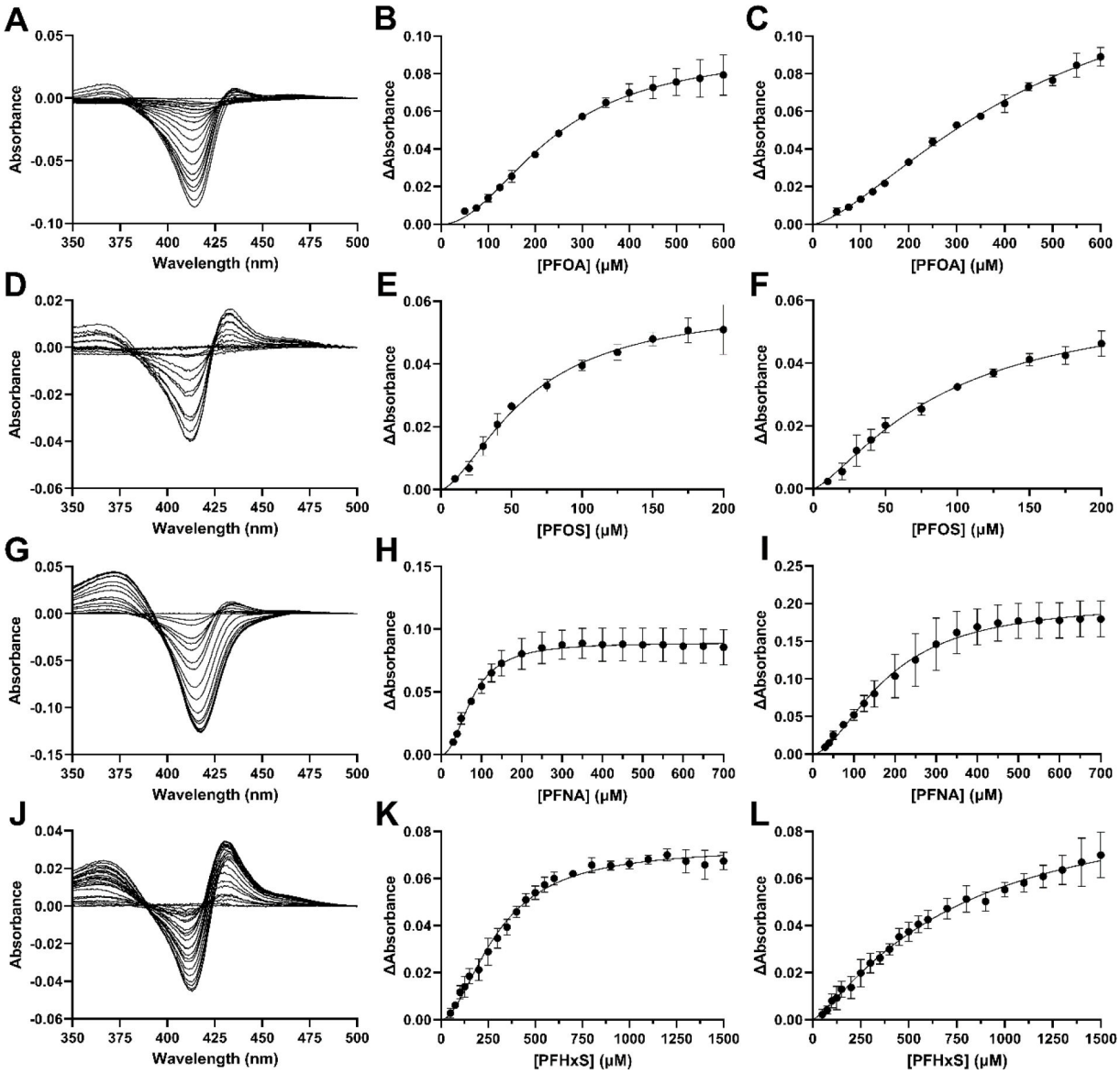


Fig. 2.

PFAS representative binding spectra for rCYP3A7 with the two corresponding binding isotherms. Representative binding spectra for PFOA, PFOS, PFNA, and PFHxS obtained with 2 μM recombinant CYP3A7 shown for each respective compound. Each binding spectra is accompanied with isotherm data from the first and second binding events at each compound concentration across all three experimental replicates ($n = 3$). (A) PFOA representative binding spectra and binding isotherms for the first (B), K_d = 237.5 μM (R^2 = 0.975), and second (C) K_d = 477.9 μM (R^2 = 0.990) binding events. (D) PFOS representative binding spectra and binding isotherms for the first (E), K_d 61.74 μM (R^2 = 0.970), and second (F) K_d = 90.39 μM (R^2 = 0.973) binding events. (G) PFNA representative binding spectra and binding isotherms for the first (H), K_d = 76.87 μM (R^2 = 0.894). and second (I) K_d = 181.2 μM (R^2 = 0.923) binding events. (J) PFHxS representative binding spectra and binding Isotherms for the first (K), K_d = 296.8 μM (R^2 = 0.977), and second (L) K_d = 731.3

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μM (R^2 = 0.962) binding events. Isotherm curves were fitted using nonlinear regression Hill equation using GraphPad Prism 9 software for MacOS (version 9.4.1).

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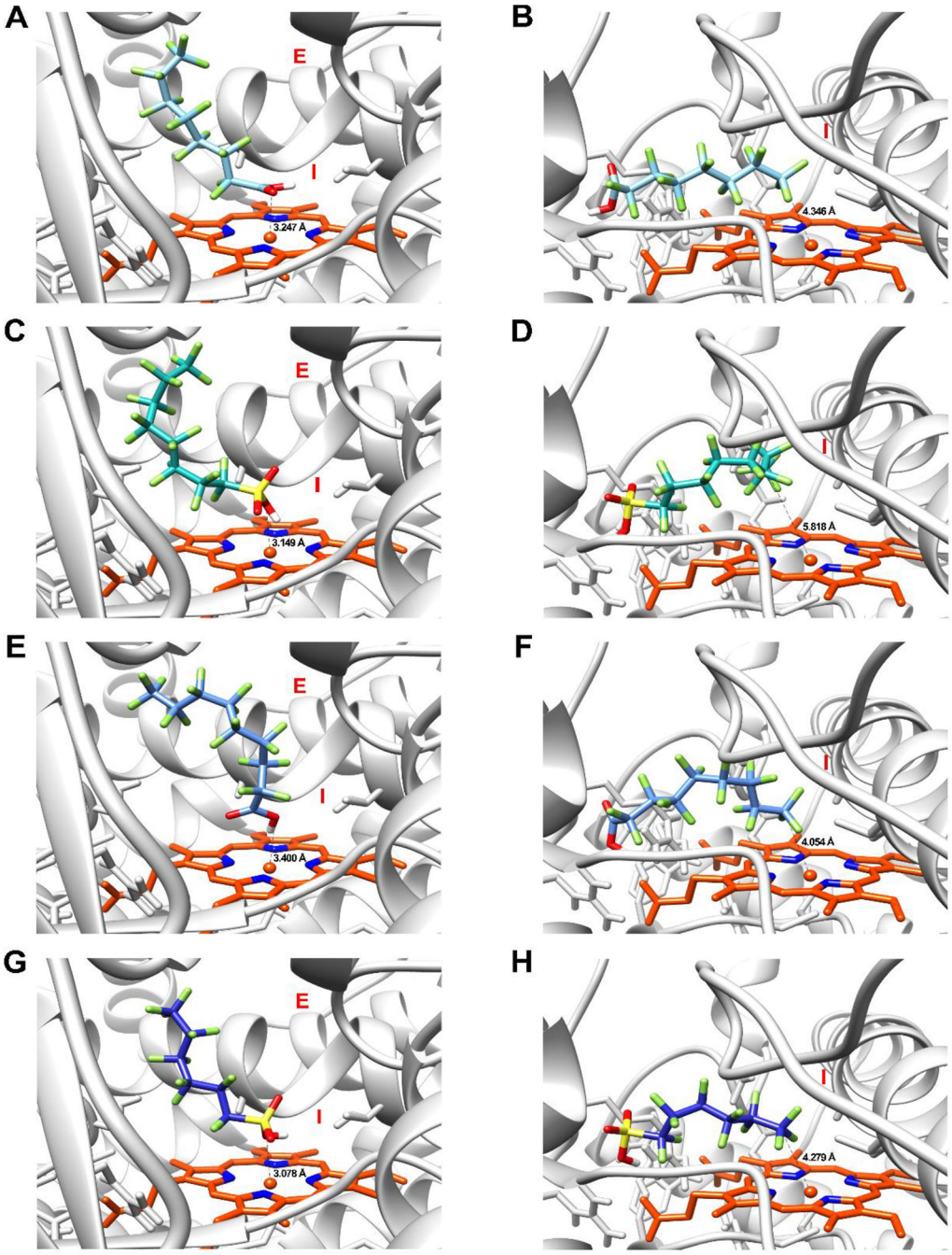


Figure 3. *In-silico* docking of PFAS with CYP3A7.

CYP3A7-active site cutaway of the docked structure of PFOA, PFOS, PFNA, and PFHxS compounds in close proximity to the heme (orange) in the active site. Reverse Type I (A, C, E, and G) coupled with Type I (B, D, F, and H) binding positions displayed for each compound in reference to CYP3A7 helices "I" and "E". Distances between each ligand and the heme iron are represented by the dashed black line and labeled in Å. Reverse Type I binding events with associated distances between the nearest heteroatom and CYP3A7 heme for PFOA (A: 3.427 Å), PFOS (C: 3.149 Å), PFNA (E: 3.400 Å) and PFHxS (G:

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3.078 Å). Type I binding events with associated distances between the nearest ω−1 carbon and CYP3A7 heme for PFOA (A: 4.346 Å), PFOS (C: 5.818 Å), PFNA (E: 4.054 Å) and PFHxS (G: 4.279 Å). Optimization and analysis performed using UCSF Chimera 1.15, AutoDockTools- 1.5.7 and AutoDock Vina software for Windows 10.

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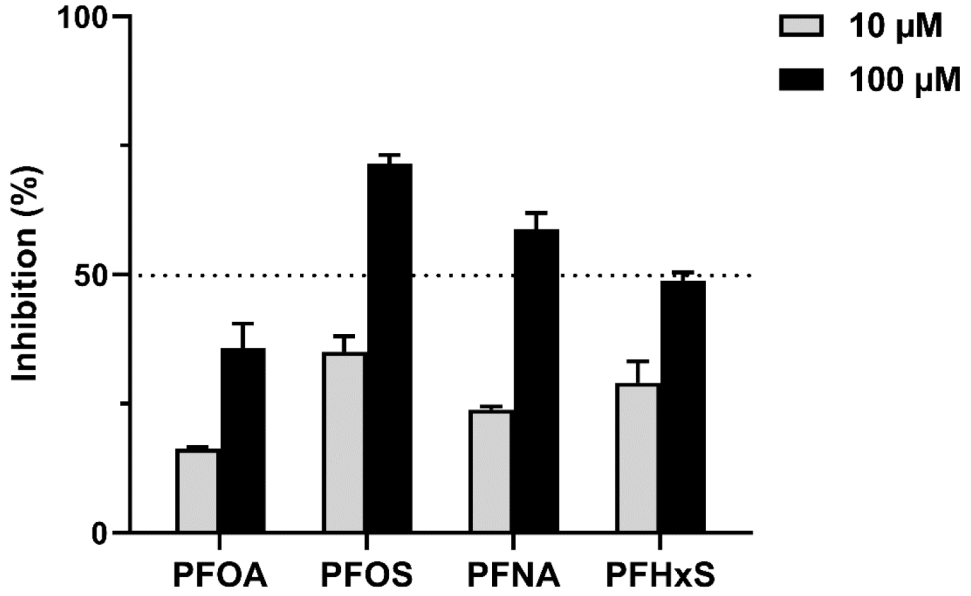


Figure 4. PFAS inhibition screening of CYP3A7 bactosome activity.

Bar chart showing mean percent inhibition of CYP3A7 oxidation of DBF by 10 and 100 µM PFOA (10 µM: 16.3% ± 0.275; 100 µM: 35.8% ± 4.76), PFOS (10 µM: 35.1% ± 2.99; 100 µM: 71.5% ± 1.66), PFNA (10 µM: 23.9% ± 0.593; 100 µM: 58.8% ± 3.15), and PFHxS (10 µM: 29.1% ± 4.19; 100 µM: 48.8% ± 1.63) against the DMSO control. Error bars represent standard deviation of triplicate values. Descriptive statistics were calculated using GraphPad Prism 9 software for MacOS (version 9.4.1).

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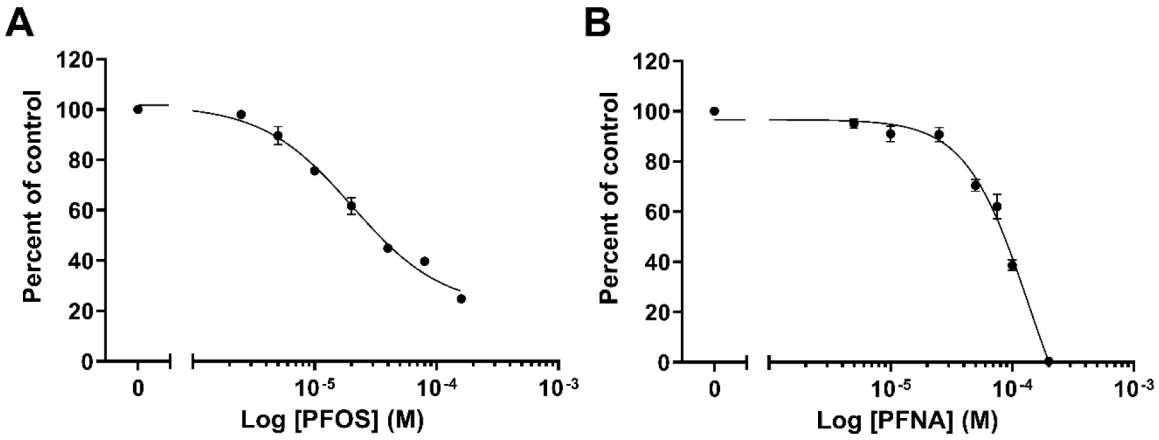


Figure 5. IC_{50} of PFOS and PFNA inhibition of CYP3A7 oxidation of DBF.

Mean values of CYP3A7 remaining activity obtained from triplicate measurements and plotted against the log of increasing concentrations of PFOS (A), IC_{50} = 20.39 μM (R^2 = 0.981), and PFNA (B), IC_{50} = 137.6 μM (R^2 = 0.978). Error bars represent the standard deviation at each concentration.

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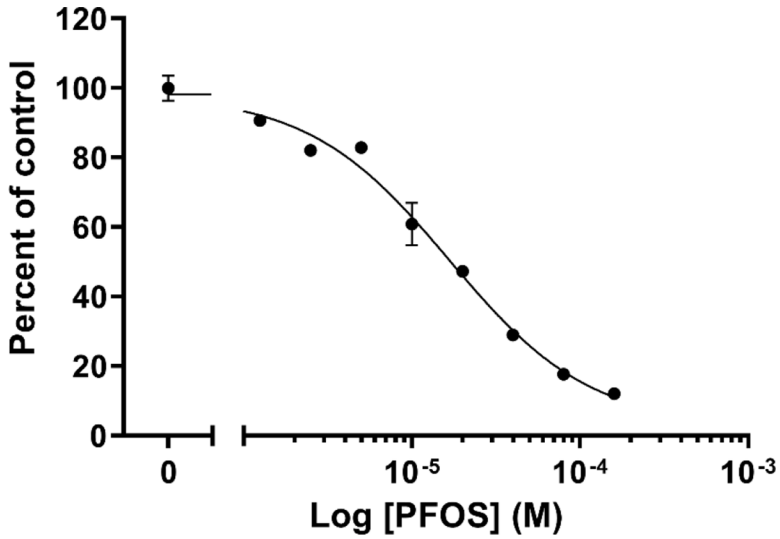


Figure 6. IC_{50} of PFOS inhibition of CYP3A7 hydroxylation of DHEA-S.

Mean values of CYP3A7 remaining activity obtained from triplicate measurements and plotted against the log of increasing concentrations of PFOS (A), IC_{50} = 16.75 μM (R^2 = 0.980). Error bars represent the standard deviation at each concentration.

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Table 1.

Binding values for each interaction between ligands and rCYP3A7.

Ligand	Binding Type	K_d, μM (95% CI)	R^2	Bmax (95% CI)	*h*
		PFAS			
PFOA	Reverse Type I	237.5 (209.7–286.2)	0.975	0.0925 (0.08417–0.1060)	2.021
	Type I	477.9 (369.9–727.7)	0.990	0.1515 (0.1253–0.2054)	1.479
PFOS	Reverse Type I	61.74 (50.72–86.71)	0.970	0.05913 (0.05260–0.07157)	1.588
	Type I	90.39 (66.38–170.0)	0.973	0.06095 (0.05074–0.08651)	1.359
PFNA	Reverse Type I	76.87 (67.68–87.41)	0.894	0.08924 (0.08508–0.09418)	2.180
	Type I	181.2 (151.4–241.4)	0.923	0.2026 (0.1827–0.2391)	1.769
PFHxS	Reverse Type I	296.8 (277.8–319 1)	0.977	0.07324 (0.07056–0.07644)	1.857
	Type I	731.3 (539.9–1283)	0.962	0.09522 (0.07990–0.1289)	1.246
		Controls			
Lopinavir *	Type I	6.482 (3.901–94.02)	0.992	0.021 (0.01668–0.06766)	1.228
Ketoconazole *	Type II	23.24 (16.65–36.36)	0.997	0.072 (0.06642–0.08085)	0.573

* Binding spectra and associated isotherms can be found in supplement Figure S1

K_d, μM (95% CI), R^2, B_{max} (95% CI), and *h* coefficients obtained for interactions between: PFAS (PFOA, PFOS, PFNA, and PFHxS; n=3) and controls (Ketoconazole [type II] and Lopinavir [type I]; n=1) and rCYP3A7 according to data from optical difference spectroscopy calculated using GraphPad Prism 9 software for MacOS (version 9.4.1).

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Table 2.

Type of binding pose and corresponding energy values and heme distances for PFAS and CYP3A7.

PFAS	Binding Type	Binding Energy (kcal/mol)	Distance (Å)
PFOA	Reverse Type I	−7.7	3.247
	Type I	−8.3	4.346
PFOS	Reverse Type I	−7.9	3.149
	Type I	−8.0	5.818
PFNA	Reverse Type I	−8.2	3.400
	Type I	−8.3	4.054
PFHxS	Reverse Type I	−7.2	3.078
	Type I	−7.5	4.279

Binding energy and distance values obtained from the molecular docking studies of PFOA, PFOS, PFNA and PFHxS compounds with CYP3A7 according to the binding poses. The distances were measured between the nearest heteroatom (reverse Type I) or ω-carbon (Type I) and the heme iron in the binding pocket.